

Semantic Twin Network Configuration

- Establish twin neural networks, $T_{context}$ and T_{focus} , where $T_{context}$ maintains contextual integrity while T_{focus} concentrates on semantic salience.
- Feed the reduced document from Phase 1 into both networks to obtain semantic projections.

Targeted Chunk Identification

- Compare outputs from $T_{context}$ and T_{focus} to discern segments that align strongly on both context and focus dimensions.
- Sections are prioritized based on the similarity of projections:

$$S_{priority} = \text{cosine_similarity}(T_{context}(d_i), T_{focus}(d_i))$$

where d_i represents a document segment and cosine similarity measures alignment between the two network outputs.

Final Document Synthesis Construct the final document synthesis by incorporating segments with high $S_{priority}$ scores, ensuring contextual coherence and specificity.

B.1.4. INTEGRATION WITH RAG SYSTEMS

The refined document content, now presenting coherent and contextually various segments, feeds into the generative module. The minimized document scope reduces computational overhead and enhances semantic coherence in generated outputs.

B.1.5. BENEFITS

- Reduced need for arbitrary chunking: By employing dynamic attention-based and semantic projections, the algorithm ensures that document segmentation is guided organically by semantic relevance rather than arbitrary heuristics.
- Improved Semantic Coherence*: Preservation of logical connections between segments enhances the quality of generative responses.
- Increased Efficiency: By narrowing down documents to their most relevant components, the generative model requires fewer resources while maintaining response quality.

This approach not only addresses the inherent limitations of traditional RAG systems when dealing with lengthy documents but also allows more refined and efficient document processing, ultimately boosting the performance of response generation.

C. Keyword Filtering

In our multi-granularity retrieval framework, we expand the keyword set by employing a keyword-neighborhood-based approach. For an keyword T_e in the current keyword set $\mathbb{E}_1$, any paper p that includes T_e will have its other keywords included in the candidate keyword set. However, this approach may introduce redundant or even noisy entities due to the following reasons:

1. Low-Relevance Co-Occurrences: Two entities with low relevance might appear together in a paper due to the specific content requirements of that paper.
2. High-Frequency Terms: Certain high-frequency terms do not effectively characterize a paper. For example, while the user-provided background might include the term “Transformer”, not all entities co-occurring with “Transformer” in other papers are relevant. This is because “Transformer” is a commonly used term in many recent publications.

To mitigate these issues, we introduce two filtering mechanisms for neighborhood-based entity expansion:

1. Co-Occurrence Threshold: An entity is supplemented only if it appears together with another entity in at least m papers.
2. Low-Frequency Selection: Inspired by the TF-IDF algorithm, we hypothesize that entities appearing frequently across the entire literature database are less representative. Thus, we retain only the n entities with the lowest frequency across all literature.

Table 4. Summarization of our used prompts.

Prompts	Place
The prompt for summary, background, and main ideas extraction, namely τ_1 .	Table 5
The prompt for entity extraction, namely τ_2 .	Table 6
The prompt for concise methods extraction, namely τ_3 .	Table 7
The prompt for background transformation.	Table 8
The prompt for brainstorming.	Table 9
The prompt for initial idea generation.	Table 10
The prompt for idea integration.	Table 11
The prompt for idea expansion.	Table 12

D. Prompt Templates

The prompt templates we provide in the appendix are summarized in Table 4.

Table 5. The prompt for background and ideas extraction, namely τ_1 .

System Message	Now you are an expert in extracting key entities from research contents. You are good at identifying the most important keywords or phrases that summarize the main topics or concepts discussed in the content.
User Message For Summary	Task Description: You are provided with the title, abstract, and introduction of a research paper. Your task is to generate a concise summary of what kind of problem does this paper aim to solve and what methods are proposed to address it. The summary should follow this format: The problem of [problem] can be addressed by [main idea/approach]. Instructions: Title: Read the title to understand the general topic of the paper. Abstract: Read the abstract to get a concise summary of the research, including the problem addressed, the methods used, and the main findings. Introduction: Read the introduction to gain a deeper understanding of the background, significance, and specific problem the paper addresses, as well as the proposed approach or solution. Based on the provided information, generate a single sentence that captures the essence of the paper, following the format specified above. Your Turn: Given the following paper information: Title: title Abstract: abstract Introduction: introduction Output: The problem of [problem] can be addressed by [main idea/approach].
User Message For Background And Main Ideas	Please read the title, abstract, and introduction of the paper again, as well as the summary you provided. Complete the following two tasks: 1. Briefly provide the two most critical motivations behind proposing these methods to address the problems. 2. Briefly provide the three most critical or innovative details of the paper that were not mentioned in your summary (It's best if these details are the new methods or techniques adopted in this paper). Output: Motivations: 1.[motivation1]. 2.[motivation2]. Details: 1.[detail1]. 2.[detail2]. 3.[detail3].